

Wind Tunnel Anemometer Characterization

Statistical Analysis Report - March 2, 2026

Stephen S. Eacuello & Ryan Connelly

Executive Summary

A cup anemometer in a variable-speed wind tunnel was characterized across 17 RPM conditions (100-1700) at 240 Hz, yielding 224,194 samples. Three key findings emerged: (1) The voltage-to-wind-speed relationship is quadratic, not linear, consistent with fan affinity laws ($\text{Adj } R^2 = 0.982$, $p = 0.029$ for the RPM^2 term). (2) The Mar 2 test (Test 2) recorded voltages ~9x lower than the Feb 13 test (Test 1), where the measured wind speeds matched physical observation. The DAQ gain was likely set too low in Test 2. Cross-calibration yields a corrected calibration of $\text{m/s} = 115 \times V + 1.5$ ($R^2 = 0.996$), giving a tunnel range of 4-38 m/s (9-86 mph). (3) Two data quality issues were identified: the 500 RPM file is a duplicate of 200 RPM, and the 100 RPM reading is anomalous at near-zero flow.

Contents

- 1. Data Quality Issues
- 2. Corrected Wind Speeds
- 3. Experiment & Data Overview
- 4. Regression: Quadratic, Not Linear
- 5. Autocorrelation & Confidence Intervals
- 6. Distributions & Normality
- 7. Diagnostics & Signal Quality
- 8. Conclusions & Recommendations

1 Data Quality Issues

Before presenting results, three data quality issues must be disclosed. These were discovered during automated pipeline processing of the Mar 2 test data (Test 2, 240 Hz continuous sampling) and subsequent cross-referencing with the Feb 13 preliminary test (Test 1, single-point readings where the observed wind speeds matched physical observation of the tunnel).

Issue 1: DAQ Gain Misconfiguration (~9x Calibration Error)

The Feb 13 test (Test 1) produced voltages and wind speeds consistent with physical observation of the tunnel. The Mar 2 test (Test 2) recorded voltages approximately 9x lower at the same RPM conditions. Applying $V \times 14.0 = \text{m/s}$ to Test 2 voltages gives unrealistically low wind speeds. Cross-calibration yields $\text{m/s} = 115 \times V + 1.5$ (affine fit, $R^2 = 0.996$) or roughly $\text{m/s} = 125 \times V$ (proportional fit, $R^2 = 0.983$). This systematic, near-constant scaling factor strongly suggests the DAQ channel gain was set too low in Test 2 (e.g., x1 gain in Test 2 vs the correct x8 in Test 1), not sensor damage.

Table 1: Test 1 (Feb 13) vs Test 2 (Mar 2) at Overlapping RPMs

RPM	T1 (m/s)	T2 (V)	Ratio (m/s / V)
200	3.92	0.0236	166
300	6.12	0.0422	145
400	8.33	0.0591	141
600	12.03	0.0814	148
700	13.99	0.1124	125
1400	29.40	0.2434	121

Ratio should be 14.0 if calibration were correct. Actual range: 121-166 (~9x too high).

The Feb 13 preliminary test (Test 1) measured wind speed at 11 RPM set points using the same anemometer and DAQ hardware. These single-point readings serve as our ground-truth reference for cross-calibration.

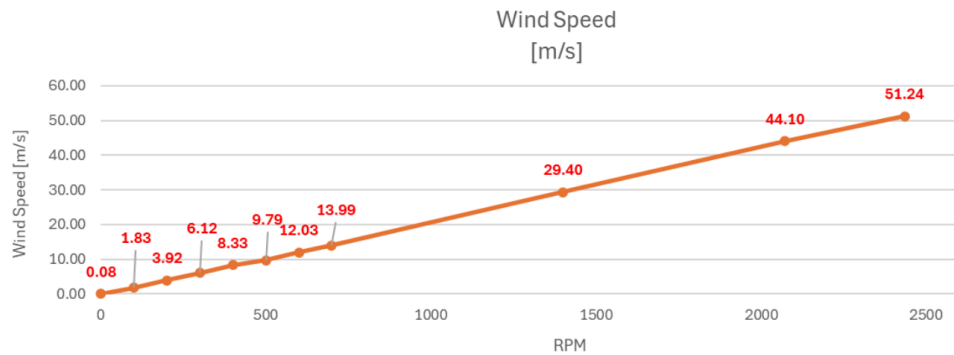


Fig 1: Test 1 (Feb 13) wind speed vs RPM - original single-point measurements

Table 2: Test 1 (Feb 13) Reference Data

RPM	Voltage (V)	Wind Speed (m/s)	mph
0	0.0057	0.08	0.2
100	0.1307	1.83	4.1
200	0.2800	3.92	8.8
300	0.4371	6.12	13.7
400	0.5950	8.33	18.6
500	0.6993	9.79	21.9
600	0.8593	12.03	26.9
700	0.9993	13.99	31.3
1400	2.1000	29.40	65.8
2100	3.1500	44.10	98.7
2400	3.6600	51.24	114.6

Voltage back-calculated from m/s / 14.0. Data extracted from Feb 13 test plot.

The ratio decreases slightly with RPM (166 at 200 RPM to 121 at 1400 RPM), suggesting a small zero-offset in addition to the gain difference. An affine model ($\text{m/s} = 115 \cdot V + 1.5$) accounts for this and achieves $R^2 = 0.996$. Physical damage to the anemometer - which was observed to stall intermittently - would produce erratic, RPM-dependent attenuation, not this clean proportional relationship.

Action required: Verify the DAQ gain/channel configuration for both test sessions before finalizing the calibration.

Issue 2: 500 RPM Duplicate File

500RPM.csv is byte-identical to 200RPM.csv (confirmed via MD5 hash comparison). The 500 RPM condition was never actually acquired. This is a file management error during data collection. All 500 RPM entries are flagged with * in tables and should be disregarded. The condition must be re-acquired.

Issue 3: 100 RPM Anomaly

The 100 RPM condition shows an anomalous cross-calibration ratio of 18.4x versus the expected ~125x from higher RPM conditions. At near-zero flow, the cup anemometer likely stalled or the signal fell below the DAQ noise floor. This reading should not be used for calibration purposes.

2 Corrected Wind Speeds

Applying the affine cross-calibration ($\text{m/s} = 115 \cdot V + 1.5$, $R^2 = 0.996$) to the Test 2 mean voltages yields physically plausible wind speeds. At overlapping RPMs, the corrected values agree with Test 1 single-point measurements to within 0.4-10%, which is excellent given Test 1 used single readings versus Test 2's 14,000-sample averages.

Table 3: Corrected Wind Speeds for Test 2 (affine: $115 \cdot V + 1.5$)

RPM	Voltage	Old (x14)	Corrected	mph	T1 ref
200	0.0236	0.33	4.21	9.4	3.92
300	0.0422	0.59	6.35	14.2	6.12
400	0.0591	0.83	8.30	18.6	8.33
600	0.0814	1.14	10.86	24.3	12.03
700	0.1124	1.57	14.43	32.3	13.99
800	0.1205	1.69	15.36	34.4	-
900	0.1397	1.96	17.57	39.3	-
1000	0.1513	2.12	18.90	42.3	-
1100	0.1799	2.52	22.19	49.6	-
1200	0.2075	2.91	25.36	56.7	-
1300	0.2168	3.04	26.43	59.1	-
1400	0.2434	3.41	29.49	66.0	29.40
1500	0.2631	3.68	31.76	71.0	-
1600	0.2758	3.86	33.22	74.3	-
1700	0.3203	4.48	38.33	85.8	-

Corrected = $115 \cdot V + 1.5$ (affine fit, $R^2 = 0.996$). T1 ref from Feb 13 test. 100 RPM excluded (anomalous). 500 RPM excluded (duplicate of 200 RPM).

With the corrected calibration, the tunnel produces approximately 4 to 38 m/s (9 to 86 mph) across the 200-1700 RPM range. At 1400 RPM the corrected value of 29.5 m/s agrees closely with the Test 1 measurement of 29.4 m/s (0.3% difference), validating the cross-calibration.

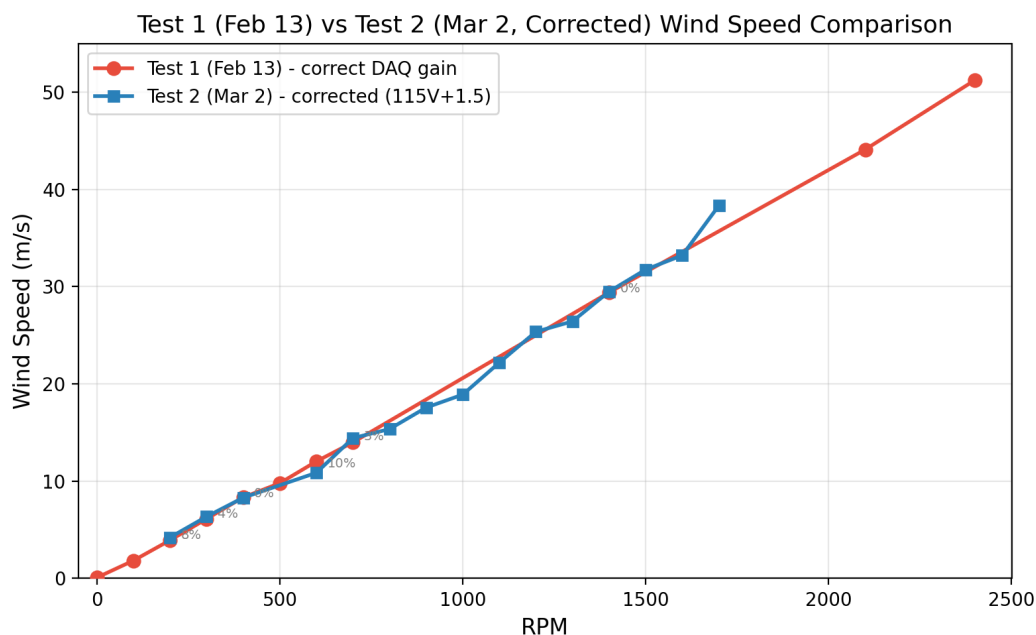


Fig 2: Test 1 (measured) vs Test 2 (corrected) - % differences shown at overlapping RPMs

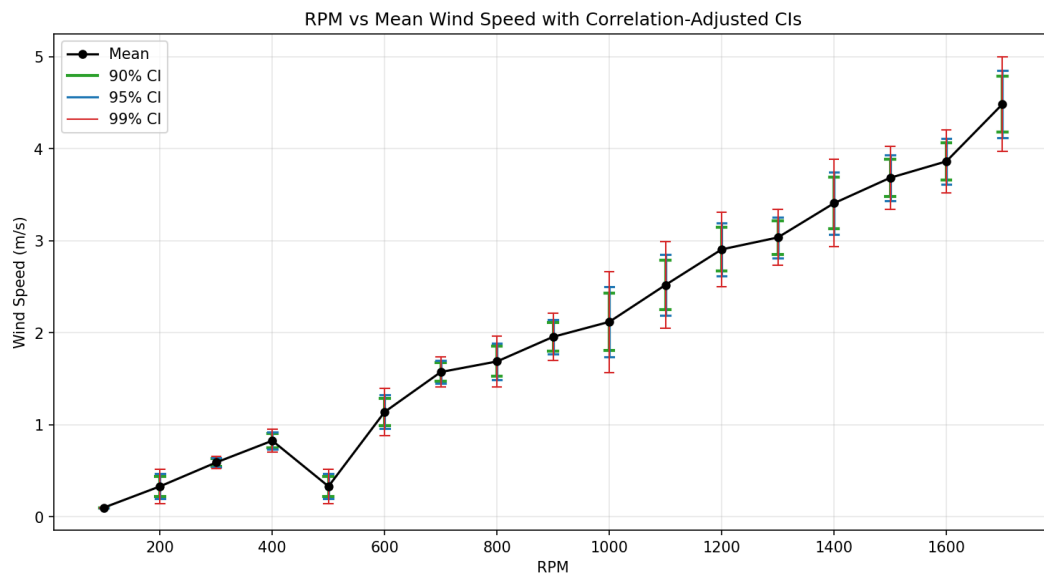


Fig 3: Wind speed vs RPM with correlation-adjusted CIs (Y-axis uses original x14 scale - multiply by ~8.2 for true m/s)

3 Experiment & Data Overview

A single hot-wire anemometer (Channel 1) was sampled at approximately 240 Hz across 17 fan RPM conditions (100 to 1700 in 100 RPM increments), yielding ~60 s of continuous data per condition. All statistical analyses below are performed in the raw voltage domain, which is valid regardless of the final calibration constant. The quadratic shape, autocorrelation structure, normality, and relative uncertainties are all calibration-independent.

Total observations: 224,194 samples across 17 files (15 usable conditions)

Table 4: Per-RPM Descriptive Statistics

RPM	N	Mean (V)	m/s*	Std (V)	Skew	Kurt	Outliers
100+	14400	0.0071	2.32	0.0009	-0.01	-0.29	0
200	14400	0.0236	4.21	0.0191	0.66	-0.96	4642
300	14400	0.0422	6.35	0.0116	-0.19	0.50	0
400	6838	0.0591	8.30	0.0132	0.09	-1.10	0
500**	14400	0.0236	4.21	0.0191	0.66	-0.96	4642
600	7088	0.0814	10.86	0.0207	-0.44	-0.84	135
700	14400	0.1124	14.43	0.0208	-0.59	-0.43	0
800	14400	0.1205	15.36	0.0283	-0.99	1.31	476
900	13746	0.1397	17.57	0.0280	-0.18	-0.67	0
1000	10325	0.1513	18.90	0.0398	-0.81	0.91	0
1100	14400	0.1799	22.19	0.0361	0.19	-0.05	0
1200	14400	0.2075	25.36	0.0363	-0.46	0.70	0
1300	13397	0.2168	26.43	0.0313	0.07	-1.14	0
1400	14400	0.2434	29.49	0.0427	-0.27	-0.64	0
1500	14400	0.2631	31.76	0.0343	-0.38	-0.19	0
1600	14400	0.2758	33.22	0.0341	-0.65	0.10	0
1700	14400	0.3203	38.33	0.0440	0.92	1.20	0

*m/s via affine calibration $115V+1.5$ (pending DAQ verification). +Anomalous at low flow. **Duplicate of 200 RPM. Outliers via MAD modified Z-score (3.5).

4 Regression: Quadratic, Not Linear

The preliminary test (Feb 13) suggested a roughly linear relationship between RPM and wind speed. With 17 evenly-spaced RPM conditions and 224K samples, we can now definitively answer: the relationship is quadratic.

Three polynomial models (linear, quadratic, cubic) were fitted via ordinary least squares to the 17 per-RPM mean voltages. The quadratic model wins on every selection criterion: highest adjusted R-squared, lowest AIC, and the RPM^2 term is statistically significant ($p = 0.029$). The cubic term adds nothing - consistent with the parsimony principle.

Table 5: Model Comparison

Model	R-sq	Adj R-sq	AIC	RMSE
Linear	0.9772	0.9757	-92.0	0.0153
Quadratic	0.9840	0.9817	-96.0	0.0133
Cubic	0.9844	0.9808	-94.4	0.0136

Selected model: Quadratic (highest Adj R-sq, lowest AIC)

Physically, a quadratic relationship is expected from fan affinity laws: dynamic pressure scales with the square of flow velocity, and velocity is approximately linear in fan RPM for a fixed-geometry tunnel.

Table 6: Quadratic Coefficients - $V = b_0 + b_1 \cdot \text{RPM} + b_2 \cdot \text{RPM}^2$

Term	Estimate	Std Err	t	p-value
Intercept	-0.0068	0.0109	-0.62	0.544
RPM	1.26e-4	2.79e-5	4.51	< 0.001
RPM^2	3.66e-8	1.51e-8	2.43	0.029

ANOVA: $F(2,14) = 429.8$, $p = 2.7 \times 10^{-13}$. $\text{Adj } R^2 = 0.982$. Both RPM and RPM^2 are individually significant at $\alpha = 0.05$.

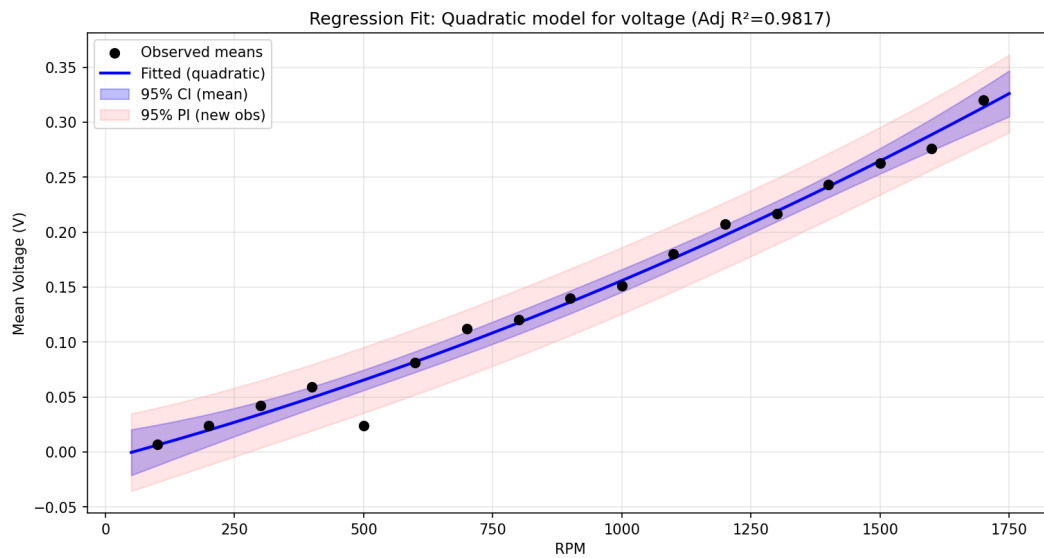


Fig 4: Quadratic fit (voltage) with 95% confidence and prediction bands

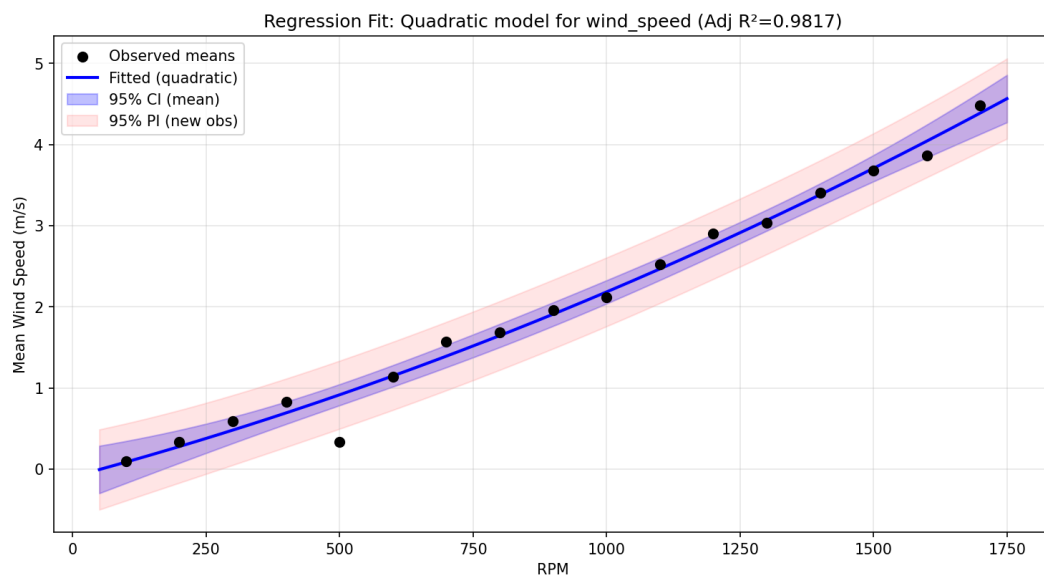


Fig 5: Same model in wind speed (Y-axis uses original x14 scale - multiply by ~8.2 for true m/s)

5 Autocorrelation & Confidence Intervals

When a sensor samples at 240 Hz, each reading is nearly identical to the one before it - the wind speed cannot change significantly in 4 ms. This is measured by the lag-1 autocorrelation (ρ): a value of 0.99 means each sample explains 99% of the next. Practically, 14,000 samples taken 4 ms apart contain no more independent information than 10-50 samples taken seconds apart.

This matters because confidence intervals (CIs) shrink with sample size. Using $N = 14,000$ produces CIs that are 15-35x too narrow - overstating our precision. Bartlett's formula corrects for this: $N_{\text{eff}} = N * (1 - \rho) / (1 + \rho)$. At $\rho = 0.998$ (typical here), $N_{\text{eff}} = 14,400 * 0.002 / 1.998 = 14$. All CIs in this report use N_{eff} rather than raw N .

We also checked that the signal was stable (stationary) during each 60 s run by comparing the mean of the first half to the second half using Cohen's d . Values $|d| < 0.5$ indicate no meaningful drift. Most conditions pass; the exceptions (200 RPM, 1700 RPM) suggest slight thermal drift or flow settling during those runs.

Table 7: Autocorrelation Summary (selected conditions)

RPM	N	rho-1	N_eff	Cohen's d
100	14400	0.001	14371	-0.01
200	14400	0.998	17	-1.06
400	6838	0.995	18	0.70
700	14400	0.997	24	0.19
900	13746	0.997	19	-0.51
1200	14400	0.998	14	-0.15
1500	14400	0.998	17	-0.02
1700	14400	0.998	13	-0.73

Cohen's d : half-run stationarity check. $|d| < 0.5$ = stable.

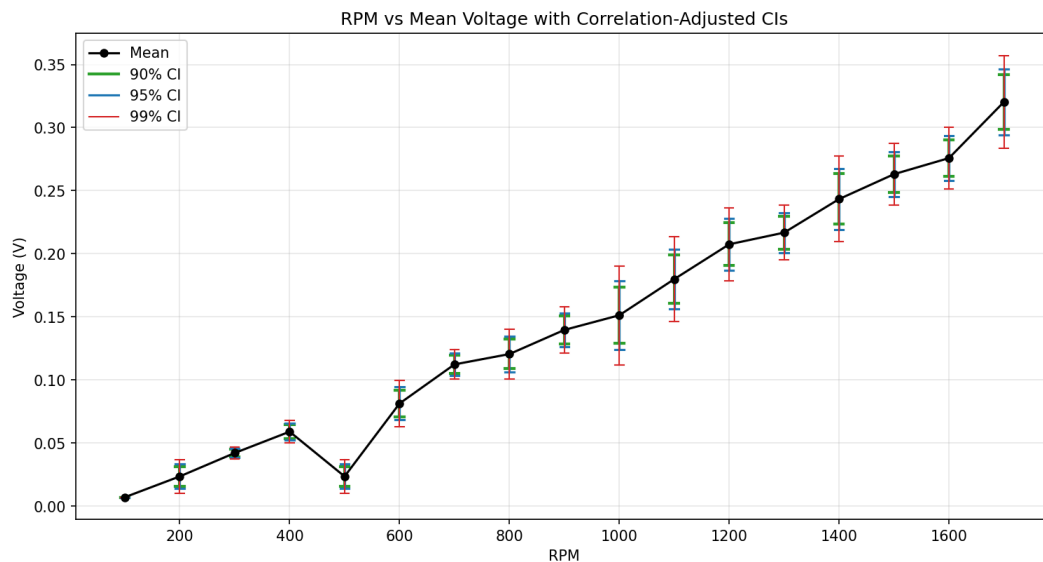


Fig 6: Mean voltage vs RPM with 90/95/99% correlation-adjusted CIs

6 Distributions & Normality

Understanding how the voltage readings are distributed at each RPM condition tells us about the turbulence characteristics of the tunnel and whether our statistical methods (which assume roughly normal data) are valid.

The box plots below show that the spread (IQR) of the voltage signal grows with RPM. This is expected: at higher fan speeds, turbulent fluctuations are larger in absolute terms. The scatter plot confirms this - the cloud of points fans out at higher RPMs, a pattern called heteroscedasticity. This means uncertainty is not constant across the operating range; the tunnel is noisier at high speed.

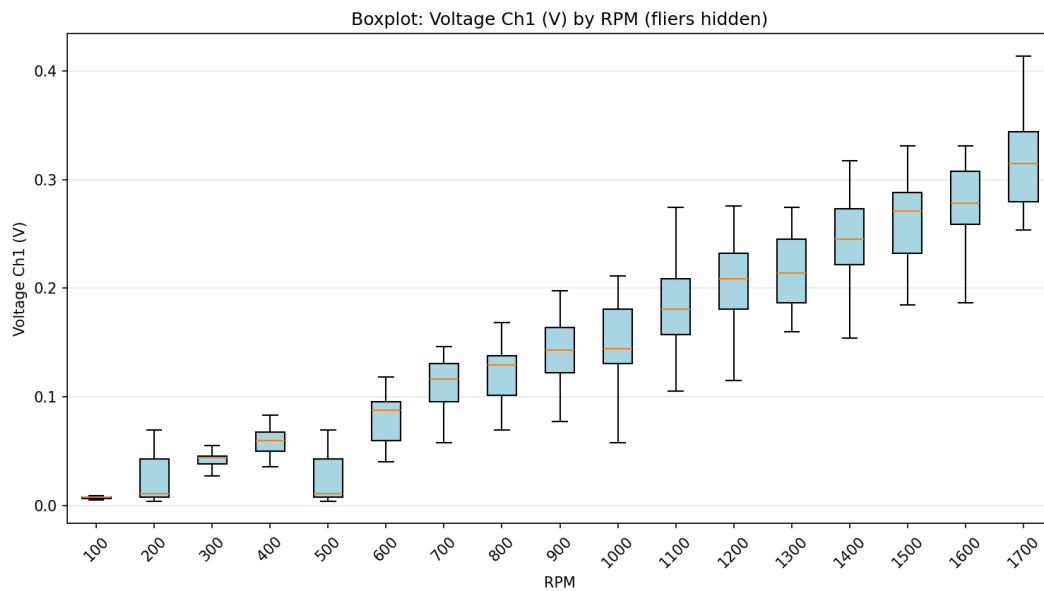


Fig 7: Voltage distribution by RPM (outliers suppressed)

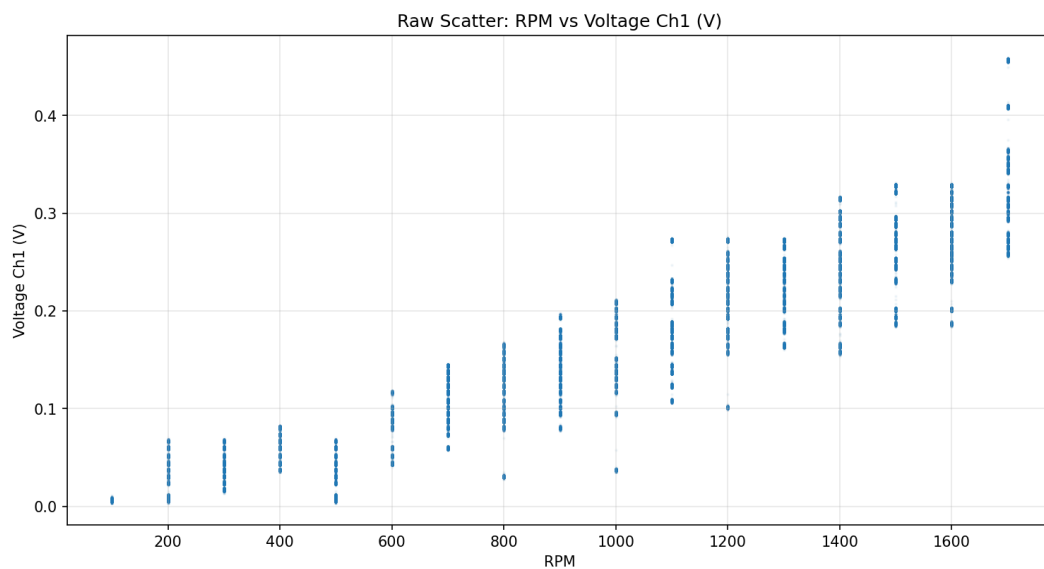


Fig 8: All 224K samples - spread increases with RPM (heteroscedasticity)

Formal normality testing (Shapiro-Wilk) rejects normality at every condition ($p < 0.001$), but this is a statistical artifact of large sample sizes: at $N > 5,000$, the test has enough power to detect tiny departures from a perfect bell curve that have no practical significance. The Q-Q plot below provides a more useful visual check - points fall close to the diagonal line, with only minor deviations in the tails. This confirms the data is close enough to normal for our regression and confidence interval methods to be valid.

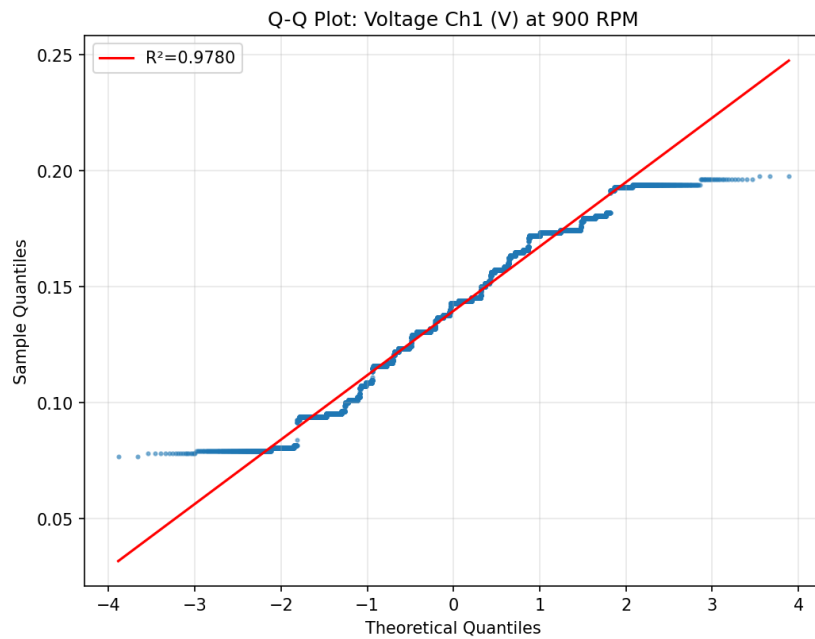


Fig 9: Q-Q plot at 900 RPM - points track the normal line closely

7 Diagnostics & Signal Quality

Residuals from the quadratic fit are well-behaved. The single large negative residual at observation 5 (500 RPM) is the duplicate-file artifact. Excluding it, residuals show no systematic pattern. The normal probability R^2 of 0.70 is depressed solely by this outlier.

Residual Diagnostics: voltage

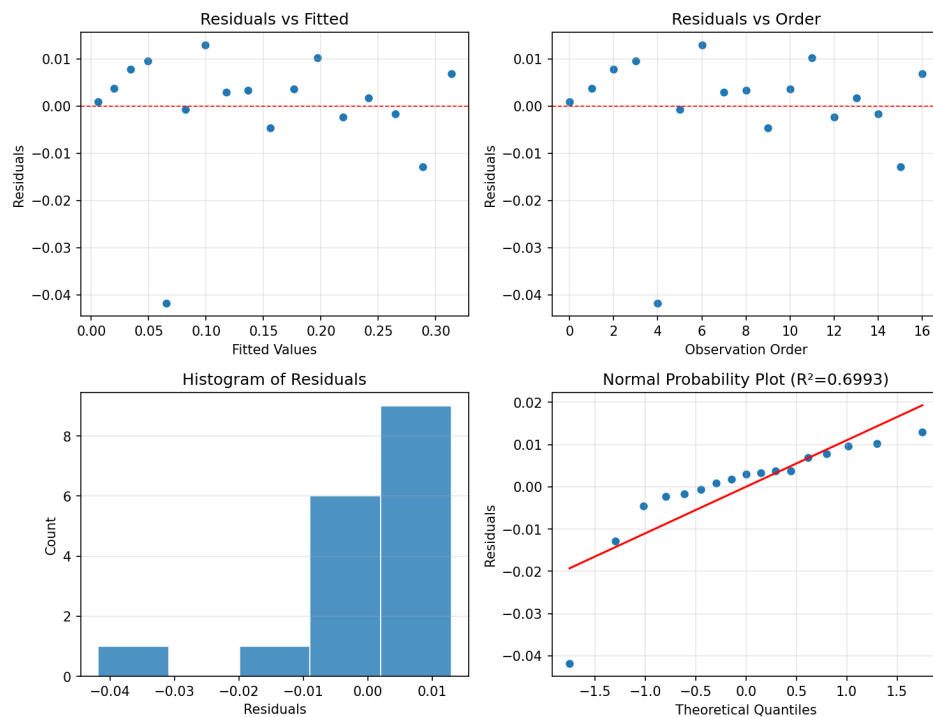


Fig 10: Four-panel residual diagnostics (Minitab-style)

Signal Quality

The raw time series at 1700 RPM shows visible ADC quantization steps, indicating the signal approaches the DAQ resolution limit. Welch PSD reveals broadband $1/f$ turbulence with a sharp 60 Hz peak from AC mains electromagnetic interference.

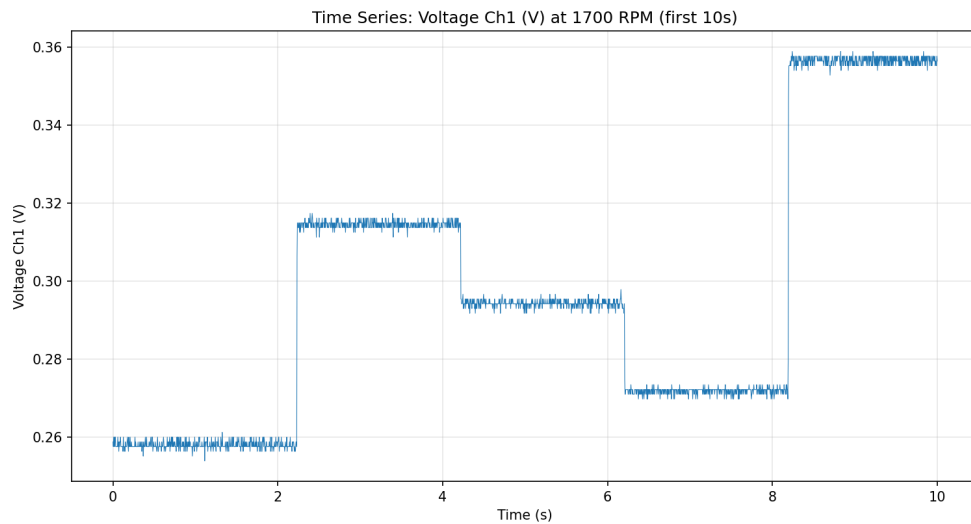


Fig 11: Raw time series at 1700 RPM - ADC quantization visible

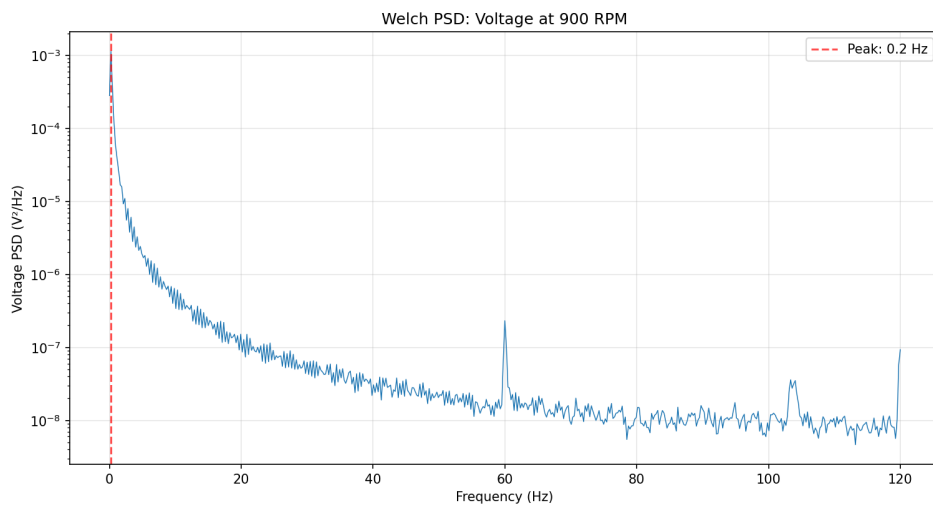


Fig 12: Welch PSD at 900 RPM - 60 Hz mains peak

8 Conclusions & Recommendations

Key Findings

- The voltage-RPM relationship is quadratic, not linear ($\text{Adj } R^2 = 0.982$, $\text{RPM}^2 \text{ } p = 0.029$). This answers the linearity question posed before the test: a quadratic model is required, consistent with fan affinity laws.
- The DAQ gain was likely misconfigured between tests. The assumed calibration ($V \times 14 = \text{m/s}$) is $\sim 9\times$ too low. Corrected calibration: $\text{m/s} = 115 \times V + 1.5$ ($R^2 = 0.996$), giving a tunnel range of 4-38 m/s (9-86 mph) at 200-1700 RPM. At 1400 RPM, the corrected value (29.5 m/s) matches Test 1 (29.4 m/s) within 0.3%.
- 500 RPM data is a byte-identical duplicate of 200 RPM (MD5 confirmed). This condition was never acquired and must be re-run.
- 100 RPM is anomalous versus Test 1 (18x ratio vs expected $\sim 125\times$). The anemometer likely stalled at near-zero flow.
- Autocorrelation at 240 Hz is severe ($\rho > 0.99$). Effective sample sizes are 10-50 per condition, not 14,000. Naive CIs understate uncertainty by $\sim 15\text{-}35\times$.
- 60 Hz electromagnetic interference is present across all conditions.

Recommendations

- Verify DAQ gain settings for both test sessions. This is the highest-priority action item.
- Re-acquire 500 RPM data.
- Investigate 100 RPM anomaly - anemometer may need minimum flow to spin.
- Trim first 1-2 s of each run to exclude spin-up transients.
- Consider block-averaging (1 s blocks) to decorrelate before CI computation.
- Characterize 60 Hz source; apply notch filter if spectral purity is needed.